

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		rate = $k[\text{CH}_3][\text{S}_2\text{O}_3^{2-}]$ award (1) for rate equation containing $k[\text{CH}_3]$ award (1) for second order rate equation		1	1	2		
		(ii)		change E_a to 81100 J mol^{-1} (1) rearrange equation to $T = -\frac{E_a}{\left(R \ln \frac{k}{A}\right)}$ (1) $T = 346 \text{ K}$ (1)	1		1 1	 3	 3	
	(b)	(i)		quenching is the sudden stopping / significant slowing of a chemical reaction in a sample (1) ensure sample composition does not change between taking sample and analysis OR during analysis (1)	1 1			2		2
		(ii)		starch	1			1		1
		(iii)		canary/bright yellow	1			1		1